

A High-Precision Identity Linking $\varphi^{12}/\sqrt{5}$, the Fibonacci Square F_{12} , and a Micro-Deviation of $1/720$

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Abstract

This note reports a high-precision numerical identity relating $\varphi^{12}/\sqrt{5}$ to the Fibonacci number $F_{12} = 144$ with a correction term of magnitude $1/720$. The deviation is of order 10^{-8} , significantly below typical experimental uncertainties in physics.

1 Statement of the Identity

Let

$$\varphi = \frac{1 + \sqrt{5}}{2}. \quad (1)$$

Define

$$X = \frac{\varphi^{12}}{\sqrt{5}}. \quad (2)$$

Numerically,

$$X = 144.001388875432 \dots \quad (3)$$

Consider the approximation:

$$144 + \frac{1}{720} = 144.001388888888 \dots \quad (4)$$

The deviation is

$$\Delta = X - \left(144 + \frac{1}{720}\right) = -1.394 \times 10^{-8}. \quad (5)$$

Thus,

$$\frac{\varphi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} + \mathcal{O}(10^{-8}). \quad (6)$$

2 Relation to Fibonacci Numbers

Using Binet's formula:

$$F_n = \frac{\varphi^n - (1 - \varphi)^n}{\sqrt{5}}. \quad (7)$$

For $n = 12$ we obtain

$$F_{12} = 144. \quad (8)$$

Hence,

$$\frac{\varphi^{12}}{\sqrt{5}} = F_{12} + \varepsilon, \quad (9)$$

where

$$\varepsilon = 0.001388875432 \dots \quad (10)$$

We observe:

$$\varepsilon \approx \frac{1}{720}. \quad (11)$$

Relative deviation:

$$\frac{|\Delta|}{X} \approx 9.68 \times 10^{-11}. \quad (12)$$

3 Precision Analysis

Absolute deviation:

$$|\Delta| \approx 1.39 \times 10^{-8}. \quad (13)$$

Relative deviation:

$$\frac{|\Delta|}{X} \approx 9.68 \times 10^{-11}. \quad (14)$$

4 Remarks

The appearance of the rational correction $1/720$ is not predicted by standard Fibonacci identities. The numerical proximity suggests a possible underlying structure, though no physical interpretation is asserted here.

5 Conclusion

We report the identity

$$\frac{\varphi^{12}}{\sqrt{5}} = 144 + \frac{1}{720} \pm 1.4 \times 10^{-8}. \quad (15)$$

The origin of the correction term $1/720$ remains an open question.

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